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SYSTEM INTEGRATION PRACTICES

Integration Models

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Lecture Objectives

- Understand three different types of integration models
- Understand the characteristics of these models
- Understand the advantages and disadvantages of each model

Overview

- Originally integration focused on bringing together a diverse set of hardware in support of a software application developed from scratch.
- Now the concept of integration is dominated by software. Organizations are increasingly focused on integrating their existing and new software to form new applications.
- EAI represents the evolution of system design and technology aimed at reducing the complexity of solving the integration problems of today.
- These options include ***sharing data*** between applications while ensuring quality and consistency, ***providing integrated front-end*** access to applications, ***bridging applications through workflow***, ***and building new applications that pull together information from existing applications*** in an innovative manner.

Integration Models

Definition of an Integration Model:

- An **integration model** is a predefined approach and configuration used to integrate software.
- It provides various options for integration approaches and configurations.

Key Attributes of an Integration Model: Each integration model may focus on one or two of the following key attributes:

Simplicity of performing the integration

- The integration model should be easy to implement, minimizing the complexity of connecting systems.
- This helps save time and reduce errors during the integration process.

Reusability of integration for different configurations

- A good integration model can be applied to multiple systems or configurations without requiring significant modifications.
- This helps save costs and effort when expanding or upgrading the system.

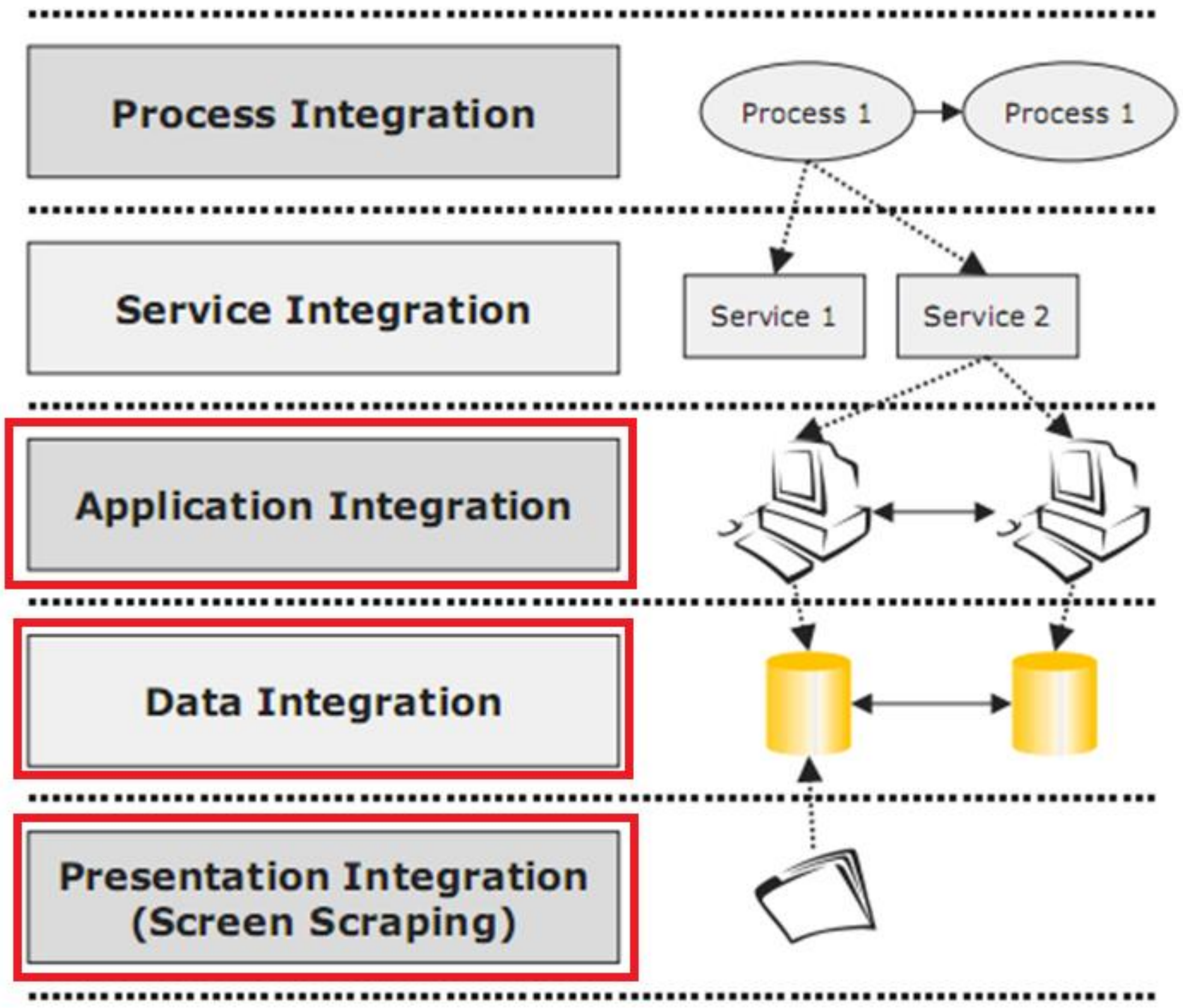
Breadth of possible approaches to integration

- Some integration models offer a wide range of approaches, giving organizations more flexibility in choosing the most suitable method.
- For example, integration can be done through APIs, data exchange, shared user interfaces, or middleware tools.

Expertise required in performing integration

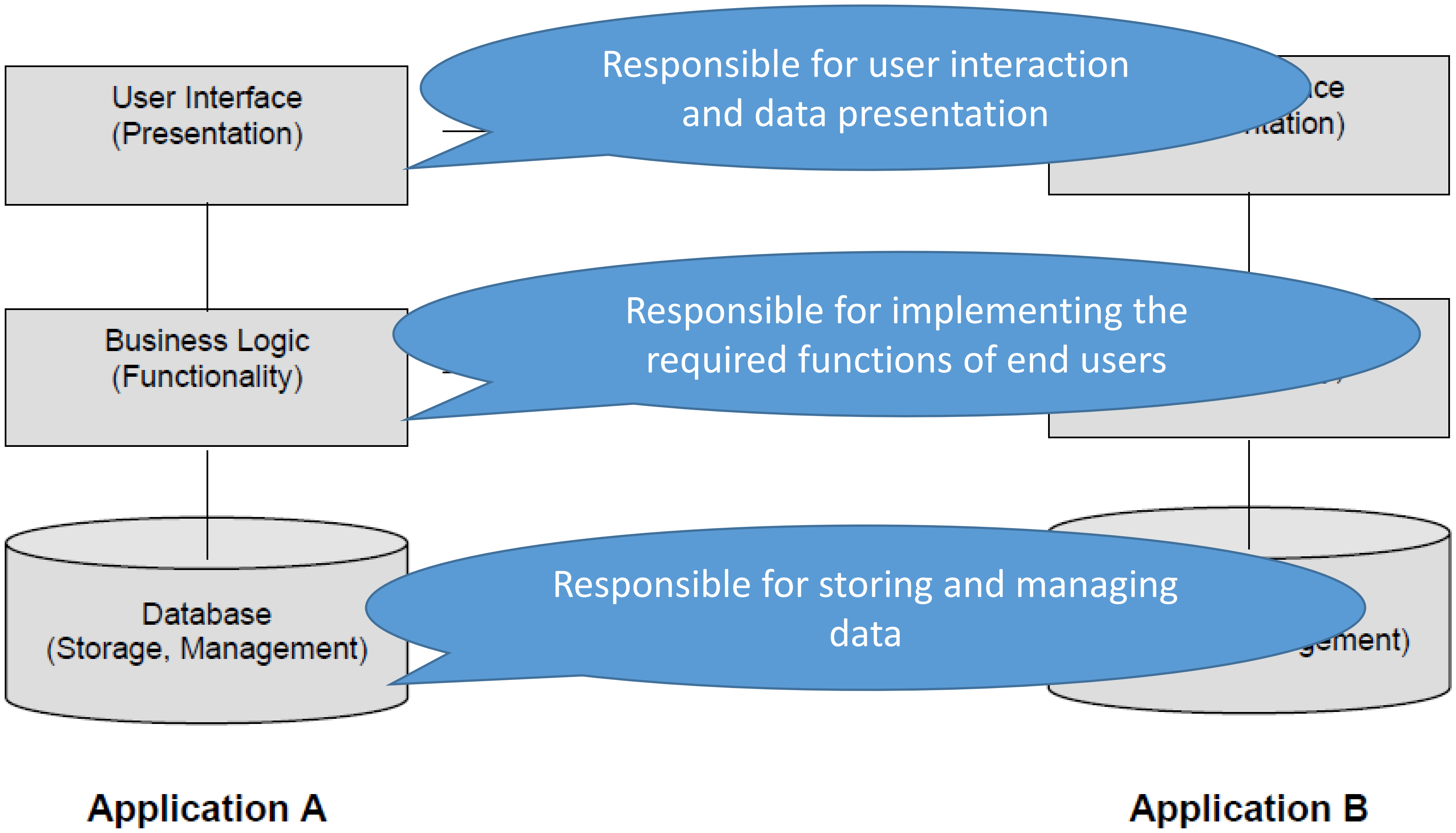
- Some models require a high level of expertise, demanding deep knowledge of systems and technologies.
- In contrast, some models are simpler, allowing technical staff with moderate skills to carry out the integration.

Models of integration



(Enterprise Architecture and Integration: Methods, Implementation, and Technologies, 2007)

Models of integration



3 Models of Integration

➤ Presentation integration:

- A presentation integration model allows the integration of new software through the existing presentations of the legacy software. This is typically used to create a new user interface but may be used to integrate with other applications

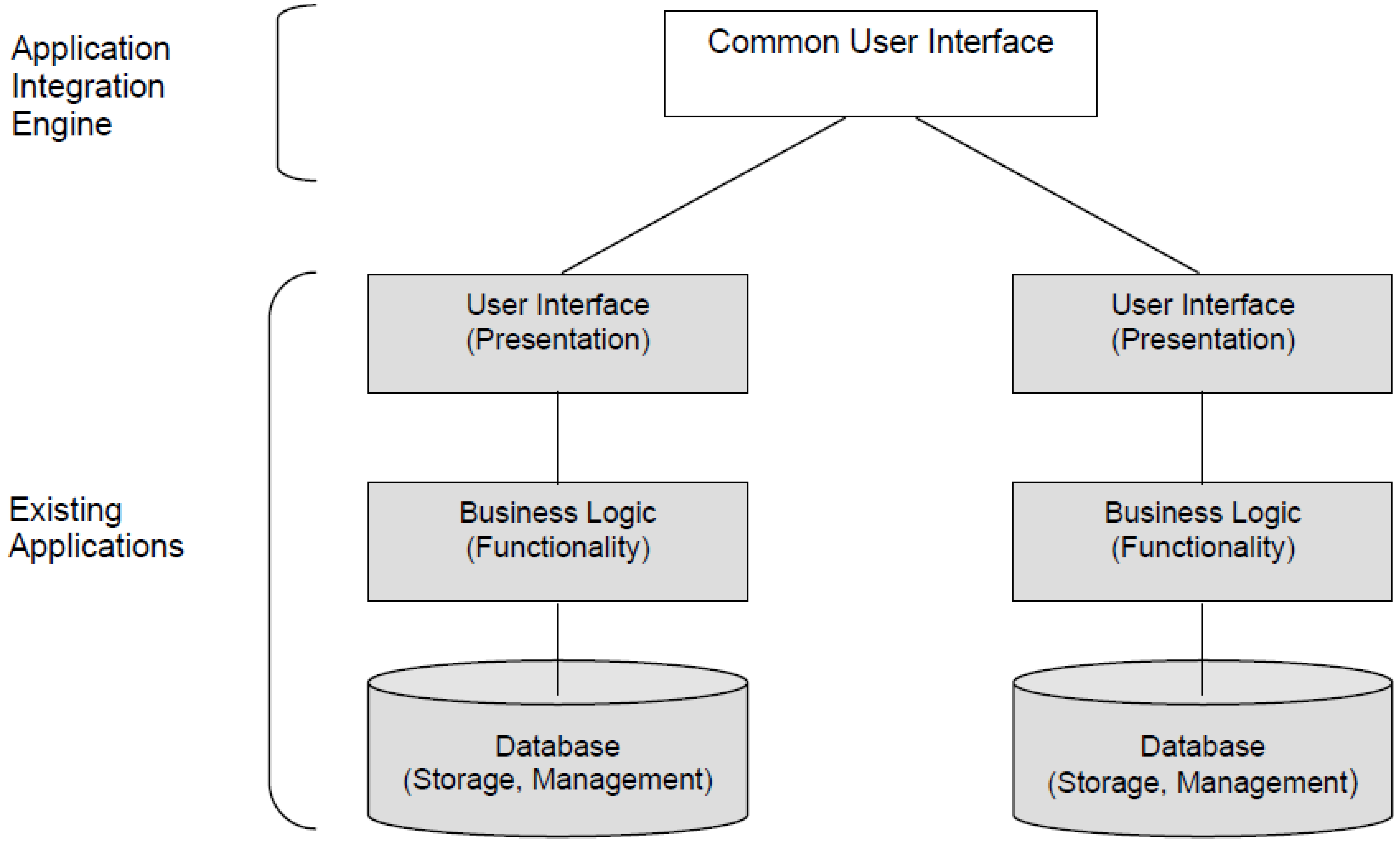
➤ Data integration

- A data integration model allows the integration of software through access to the data that is created, managed, and stored by the software typically for the purposes of reusing or synchronizing data across applications.

➤ Functional integration:

- A functional integration model allows the integration of software for the purpose of invoking existing functionality from other new or existing applications.
- The integration is done through interfaces to the software.

Presentation Integration



Integration Methods in Presentation Integration

1. Screen Scraping

Description: Screen Scraping is a technique used to extract data from the user interface (UI) of a legacy system when there is no available API or direct access to the database.
Instead of retrieving data from the backend, the new software “looks at” the data displayed on the screen just like a real user would.

How it works:

- **Observing the display interface:** The new system captures or reads the screen from the legacy application.
- **Recognizing data:**
  - If the data is displayed as text: the tool can directly read it from the control/UI element.
  - If the data is displayed as an image (bitmap, scan, etc.): Optical Character Recognition (OCR) is used to identify the characters.
- **Simulating user actions:** Scripts or software robots input data, click buttons, switch tabs, etc., mimicking manual user interactions.

Integration Methods in Presentation Integration

1. **Screen Scraping**

❑**Supporting Technologies:**

- **OCR (Optical Character Recognition):**
  - Tools such as Tesseract OCR, ABBYY FineReader, and Google Cloud Vision API can recognize text from images.
- **UI Automation & Scripting Tools:**
  - Tools like AutoIt, Selenium, UiPath, Blue Prism, and Power Automate are used to automate clicking, data entry, and data copying from the user interface.
- …

Integration Methods in Presentation Integration

1. Screen Scraping

Advantages

- No need to modify or directly access the legacy system.
- Quick to implement when the system has no available API.
- Can be applied to various types of applications (mainframe, web, desktop).

Disadvantages

- **Low reliability:** Even small interface changes (color, button position, layout) can break the script.
- **Poor performance:** Compared to APIs or database queries, screen scraping is slower because it operates through the UI.
- **Limited scalability:** Suitable for small tasks but not recommended for large systems.
- **OCR inaccuracies:** If the interface is blurry or complex, OCR may misread characters.

Practical Applications

- **RPA (Robotic Process Automation):** Automating data entry from legacy systems to new ERP systems.
- **Data Migration:** Extracting data from systems without APIs to transfer into new databases.
- **Test Automation:** Simulating user behavior for interface testing.

Integration Methods in Presentation Integration

2. UI API

Definition: is a method of integration at the user interface layer. However, instead of "reading the screen" (as in screen scraping), the new system directly calls the APIs exposed by the legacy application to interact with its UI.

These are usually **UI-level APIs** or **automation APIs** provided by the application itself, allowing external programs to access and control the interface (*perform actions, retrieve data, fill in forms, click buttons, etc.*).

How it Works

- The legacy application provides built-in **UI APIs or Automation APIs** (e.g., Windows Automation API, Java Accessibility API, COM-based UI API, Web UI APIs).
- The middleware or new system directly invokes these API functions to:
  - **Read data** from UI controls (textbox, table, label, etc.).
  - **Perform operations:** click buttons, enter data, select menus.
- **No OCR required:** since data is retrieved directly from UI controls, it is much more accurate than screen scraping.
- The extracted results are then sent to the new system through processing workflows such as **ETL**, **workflow**, or **integration middleware**.

Integration Methods in Presentation Integration

2. UI API

Technologies & Supporting Tools

- **Windows UI Automation API (Microsoft UIA):** Allows access to controls, properties, and events in Windows applications.
- **Java Accessibility API:** Enables external tools to control Java Swing/AWT applications.
- **COM/OLE Automation:** For example, Microsoft Office API allows manipulating Word and Excel through code.
- **Web UI API:**
  - DOM API and JavaScript API enable direct interaction with web elements.
  - Selenium, Puppeteer, and Playwright utilize the browser’s UI API.
- **RPA Tools:** UiPath, Blue Prism, and Automation Anywhere often combine UI APIs to improve accuracy instead of relying solely on screen scraping.

Integration Methods in Presentation Integration

2. UI API

Technologies & Supporting Tools

```
using System.Windows.Automation;
// Tìm cửa sổ Notepad
AutomationElement notepad = AutomationElement.RootElement.FindFirst(
    TreeScope.Children,new PropertyCondition(AutomationElement.NameProperty, "Untitled - Notepad")
//AutomationElement.RootElement: đại diện cho desktop (toàn bộ cây UI).
// FindFirst(reeScope.Children, ...): tìm trong danh sách các cửa sổ con trực tiếp của desktop
//PropertyCondition(AutomationElement.NameProperty, "Untitled - Notepad"): điều kiện là tên cửa sổ bằng "Untitled - Notepad".
);
// Tìm nút "Edit" menu
AutomationElement editMenu = notepad.FindFirst(
    TreeScope.Descendants,
    new PropertyCondition(AutomationElement.NameProperty, "Edit")
);
// Click nút
InvokePattern invokePattern = editMenu.GetCurrentPattern(InvokePattern.Pattern) as InvokePattern;
invokePattern.Invoke();
```

Integration Methods in Presentation Integration

2. UI API

Advantages:

- **High accuracy:** Retrieves data directly from controls, unaffected by font, color, or OCR issues.
- **More stable than screen scraping:** Minor UI changes (color, style) do not affect integration, as long as control IDs and APIs remain the same.
- **Powerful automation:** Can control the application like a real user via APIs.
- **Scalability:** Easily integrated into RPA or workflow solutions.

Disadvantages:

- **Dependent on the legacy application:** Cannot be applied if the application does not provide a UI API.
- **Challenging initial setup:** Requires a good understanding of the application's API (often poorly documented).
- **Limited compatibility:** Each platform (Windows, Web, Java, .NET, SAP, etc.) has its own API → separate adapters are needed.

Integration Methods in Presentation Integration

2. UI API

Practical Applications:

- **Office Integration:** Use Excel COM API to automatically read/write data instead of manual copying.
- **ERP Integration:** Some ERP systems (SAP, Oracle Forms) provide UI-level APIs to read data.
- **Web Automation:** Access the browser’s DOM API to accurately extract data from web applications.
- **RPA:** Combine UI APIs with scripts for fast and reliable operations on legacy systems.

Integration Methods in Presentation Integration

Comparison: Screen Scraping vs UI API

Aspect	Screen Scraping	UI API
Definition	Extracts data by reading the display screen of a legacy system when no API is available.	Integrates at the UI layer by directly calling APIs provided by the legacy system.
Data Access	Observes UI visually; may require OCR for images.	Accesses UI controls directly via API; no OCR needed.
Accuracy	Lower accuracy; sensitive to UI changes, font, color, layout.	High accuracy; minor UI changes don't affect data if control IDs/API remain the same.
Performance	Slower due to interaction through UI.	Faster; direct API calls.
Stability	Low stability; small UI changes can break scripts.	More stable; resilient to minor visual/UI changes.
Automation	Limited automation; mimics user actions via scripts.	Strong automation; can fully control application like a real user through API.
Scalability & Maintenance	Hard to scale; maintenance heavy for large systems.	Easier to scale and integrate into RPA/workflows.
Dependency	Independent of application APIs; works even without APIs.	Depends on the application providing UI APIs.

Characteristics of Presentation Integration

- Databases are independent – No coupling!
- Information presentation is through application API's
- Databases may have inconsistent information
- Primarily a display (i.e., read only)
- Examples of Presentation Integration models:
  - *Executive dashboards*
  - *Operational status displays*

When to Use It?

When to Use Presentation Integration?

- When the legacy system **has no API or database access**.
- When replacing or upgrading the legacy system is too costly or risky.
- When **creating a new user interface** while keeping the old system intact.
- When **quick integration** is needed without disrupting the existing system.

When Not to Use Presentation Integration?

- When the legacy system has an API or database access → **Data Integration or Services Integration** would be better options.
- When high-performance and complex integrations are required.
- When the legacy system’s UI **frequently changes**, which could disrupt the integration.

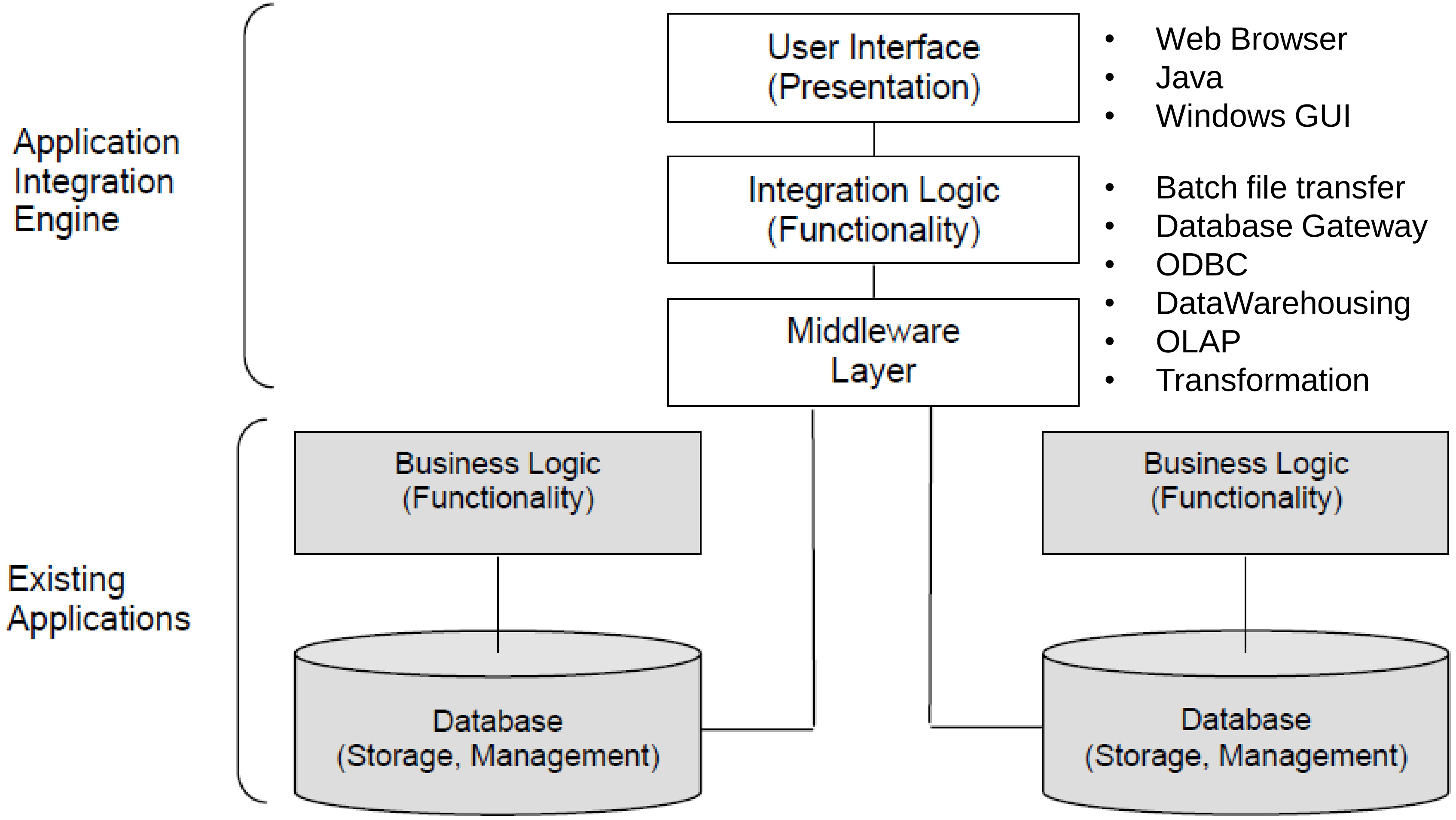
Presentation Model Advantages

- Low risk, low cost
- Technology is available and stable
- Easiest to implement of all the models
- Information presentation often meets user need for information
- Quickly implement
- Legacy applications unaffected

Presentation Model DisAdvantages

- Presentation integration occurs only at the user interface level. Therefore, only the data and interactions defined in the legacy presentations can be accessed.
- Presentation integration can have performance bottlenecks because it adds an extra layer of software to the existing application.
- The underlying data and logic of the existing application cannot be accessed.
- Maintenance can be difficult:
  - Changes in underlying databases may affect GUI
  - Data sources may not be able to change to meet requirements

Data Integration Model



Data Integration Model

Overview: Data Integration is the process of combining data from multiple sources into a unified system for analysis, reporting, or other applications.

The data integration model typically accesses the database management system (DBMS) or the data structures of an application directly. Integration can be as simple as retrieving data from an existing DBMS or as complex as handling data from custom databases or application-managed files.

Data Integration Model

Data Integration Examples Across Systems

1. **Integrating Multiple Databases**
   - Combine customer, sales, and financial databases to produce consolidated financial reports.
2. **Integrating Data for Business Intelligence (BI)**
   - Merge data from CRM, ERP, and online sales into BI tools (e.g., Tableau, Power BI) for detailed analysis and strategic decision-making.
3. **Cloud & On-Premise Integration**
   - Integrate Salesforce (cloud CRM) with on-premise ERP systems to provide a complete view of customers and business operations.
4. **Marketing Campaign Data Integration**
   - Combine online advertising platforms (Google Ads, Facebook Ads) with CRM data to track campaign performance and optimize strategies.
5. **Financial Industry Data Integration**
   - Merge payment systems, banking transactions, and stock market data for cash flow monitoring, risk analysis, and investment strategy optimization.
6. **IoT Sensor Data Integration**
   - Integrate IoT sensor data (temperature, humidity, motion) with building management or alert systems for automated decisions and timely notifications.

Data Integration Model

Technologies and Methods for Data Integration

Batch File Transfer

- This method often uses FTP, SFTP, or ETL (Extract, Transform, Load) tools to transfer data from one system to another on a scheduled basis.
- **Advantages**: Simple, easy to implement.
- **Disadvantages**: Does not support real-time processing, high latency.

Data Integration Model

Technologies and Methods for Data Integration

Open Database Connectivity (ODBC): is a standard API that allows applications to access databases regardless of the database management system (SQL Server, MySQL, PostgreSQL, etc.).

It is widely used in distributed systems and applications that need to query data from multiple sources.

Advantages: High flexibility, supports multiple database management systems.

Disadvantages: Performance may be low when querying large datasets or using it over a network.

Data Integration Model

Technologies and Methods for Data Integration

Database Access Middleware: Middleware tools act as intermediaries to connect and exchange data between different systems without requiring changes to their internal data structures.

Examples: JDBC (Java Database Connectivity), ADO.NET (for .NET applications).

Advantages: Provides abstraction for data access, making it easier to integrate multiple data sources.

Disadvantages: May require complex configurations, affecting performance.

Data Transformation

Data Integration Model

Technologies and Methods for Data Integration

Data Transformation:

- This process converts data from the source format to a format suitable for the target system.
- It may include data cleansing, data type conversion, merging, or splitting data.
- Common tools: SQL Server Integration Services (SSIS), Apache Nifi, Talend, Informatica.

Advantages: Supports powerful data processing and optimizes integration workflows.

Disadvantages: Requires advanced technical skills and can be costly to implement.

Emerging Trends in Data Integration

- **Real-time Data Integration**: Modern systems increasingly require data to be updated instantly, using technologies like Apache Kafka and Change Data Capture (CDC).
- **Cloud Data Integration**: Many organizations leverage services like AWS Glue and Google Dataflow to process and integrate data in the cloud.
- **AI & Machine Learning in Data Integration**: These technologies help process data more intelligently by automatically detecting and correcting data errors.
- **Data Virtualization**: Enables querying and combining data from multiple sources without physically moving the data.

Characteristics of the Data Integration

- Data is accessed directly from all data sources
- Middleware technology is used to facilitate data services –increasingly coupled
- Business intelligence application is often used to retrieve/update and present information
- Information can be updated in both data sources
- Existing application interfaces are not used

When is the Data Integration Model Appropriate?

- **When there are multiple data sources**: It helps combine and synchronize data from different sources into a unified system.
- **When real-time data processing is required**: It is suitable when data needs to be retrieved and updated instantly from different systems.
- **When reporting and aggregation analysis is needed**: It supports creating reports and analyses from integrated data across multiple sources.
- **When ensuring data consistency and synchronization**: It ensures accurate and consistent information across systems.
- **When reducing the complexity of data access**: It simplifies the process of retrieving and processing data from different sources.

When not using the Data Integration Model Appropriate?

- **Simple or less complex data:** If working with a single, simple data source, integration may not be necessary.
- **High implementation costs:** If the cost of implementation exceeds the benefits, using this model is not a good choice.
- **Need for high flexibility and frequent changes**: If the system requires frequent changes and flexibility, the integration model may not be suitable.
- **Sensitive data with strict security requirements**: If dealing with sensitive data that requires high security, integration may increase security risks.
- **Lack of infrastructure or technical resources:** If there are insufficient resources or infrastructure, it will be difficult to maintain and operate the integration model effectively.

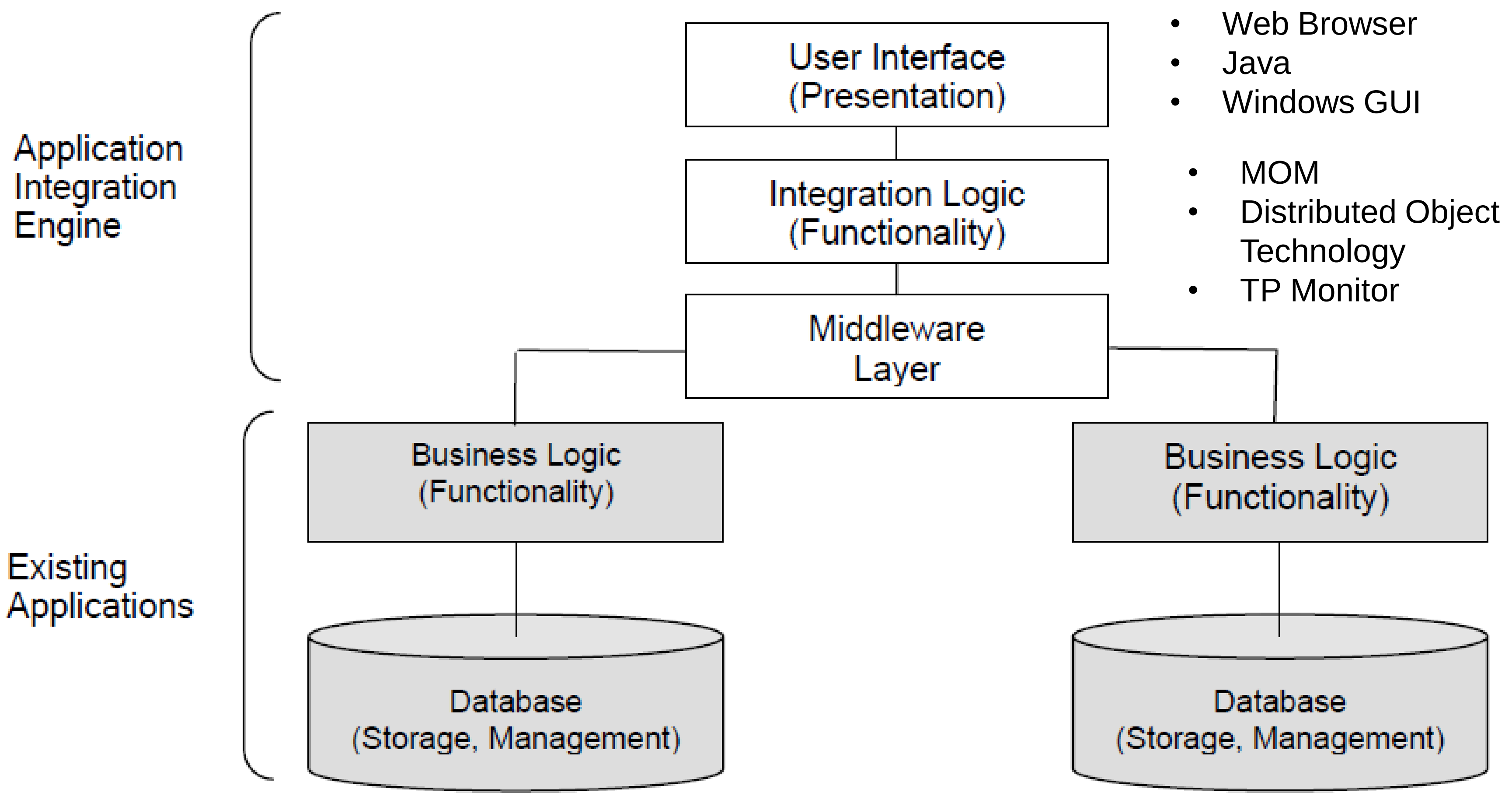
Data Integration Model Advantages

- Greater flexibility than the presentation integration model.
- Can access all data elements in the DB.
- Increase customization over presentation model
- Can update data if necessary
- Simplifies access to data sources.
- Availability of staff and technology
- Allow data to be reused across other application
- Inexpensive and proven technology
- Legacy applications unaffected

Data Integration Model DisAdvantages

- Maintenance can be difficult
- Data may be inconsistent
- The need to rewrite the business logic may appear to be a minor issue
- Each integration is tied to a data model. If a data model changes, the integration may break, making data integration sensitive to change. Because systems tend to be evolutionary this change can require significant effort to maintain the integration.

Functional Integration Model



Functional Integration Model

- This model focuses on **integration at the business logic level**, not just at the data or presentation layer.
- Integration happens **within the application’s code**, usually through a **published API** that other systems can use.
- Compared to data integration and presentation integration, **functional integration is more flexible** because it allows direct interaction with the application’s business functions.

Three Approaches to Functional Integration

1. Data Consistency

- Ensures that when data is updated in one system, the changes are also reflected in other integrated systems.
- Example: Updating customer information in a CRM system should also update the same customer data in the Sales System.

2. Multistep Process

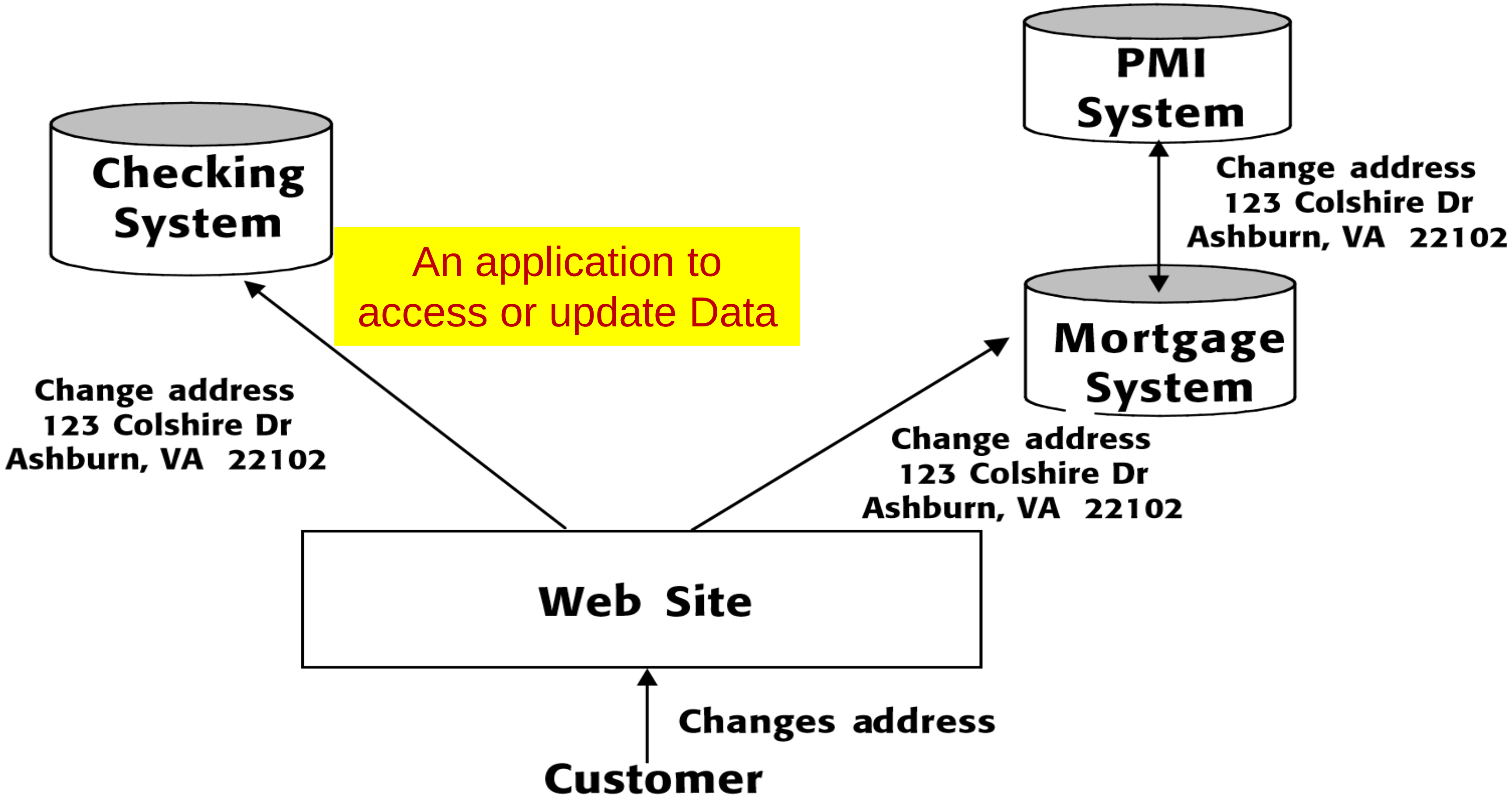
- Combines multiple actions across different systems to complete a single business process.
- Example: An online order goes through several systems → check inventory in the Inventory System → calculate shipping cost in the Logistics System → process payment in the Payment System.

3. Plug-and-Play Components

- Builds reusable components or interfaces that simplify the development of new applications.
- Example: A payment API that can be reused across different platforms (e-commerce website, mobile app, ERP system).

Functional Integration Model

Data consistency. The coordination of information updates from one or more sources across integrated applications.



Data integration should ensure data consistency across applications

Functional Integration Model

Multistep process. A coordinated set of actions executed across integrated applications.

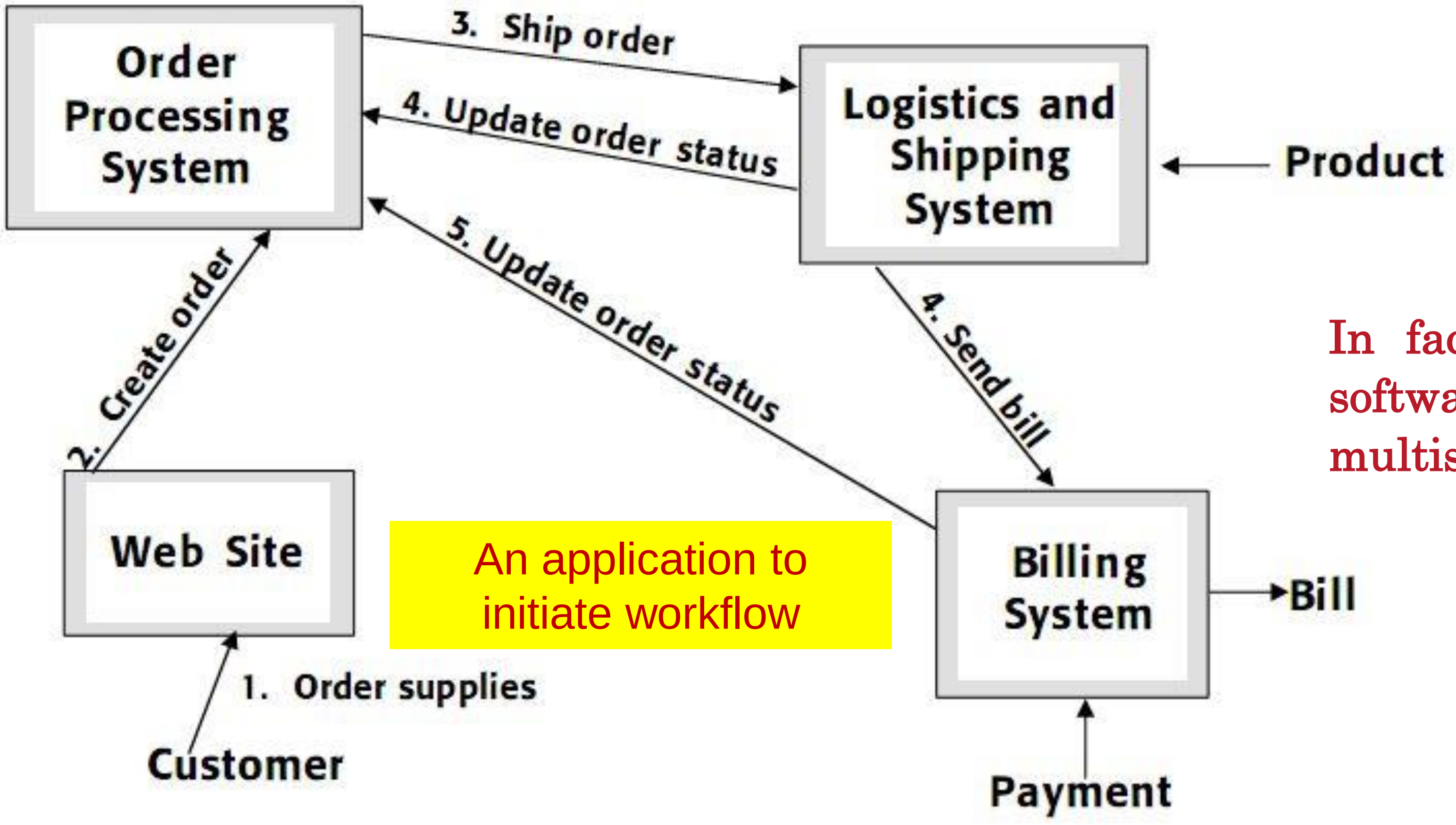


Figure 2.5 Multistep process integration is a form of business automation.

In fact, workflow software is one type of software that can be used to implement a multistep process integration.

Functional Integration Model

Components Integration. The creation of reusable interfaces across applications that simplify construction of new applications.

- The interfaces are formed using a consistent set of interface rules.
- The definitions of the actions that can be performed are consistently applied.

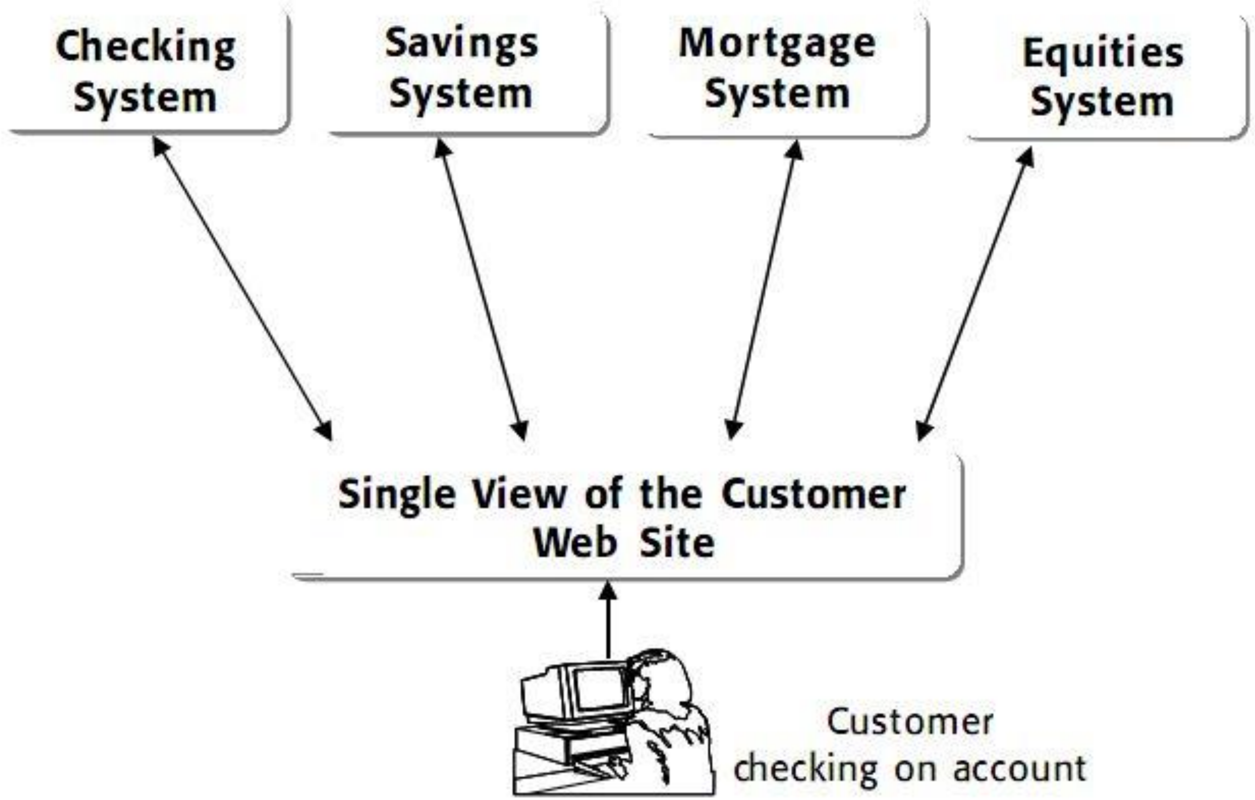


Figure 2.6 Component integration is the most practical way to build and integrate applications.

When is the Functional Model Appropriate?

The functional integration model is suitable when:

1. There is a need to integrate applications with complex business logic

- When multiple systems must work together to execute related workflows.
- Example: An e-commerce platform needs to connect with inventory management, payment processing, and customer service systems in an automated workflow.

2. High scalability and flexibility are required

- When an architecture needs to be easily modified or expanded without affecting the existing system.
- Example: A company wants to add multiple payment service providers without making major changes to the codebase.

3. Different systems with varying technologies need to be integrated

- When applications cannot share a common database but still need to work together.
- Example: A .NET-based application needs to communicate with a Java-based application via an API.

4. Automation of workflows across multiple systems is required

- When multiple applications must coordinate to complete a common task.
- Example: An insurance company needs to connect its customer registration system with identity verification and risk assessment systems before approving a policy.

When not to use the Functional Model Appropriate?

1. **Real-time performance is a critical requirement**
   - Since functional integration often requires multiple API calls or intermediary services, it may introduce latency compared to direct database integration.
   - Example: In financial systems that require real-time transaction processing, data integration might be a better choice.
2. **The system lacks APIs or the source code cannot be modified**
   - If an application does not support APIs or its code cannot be modified, implementing functional integration is difficult.
   - Example: An old accounting software without API support cannot be integrated using this approach.
3. **Development and maintenance costs are too high**
   - If the cost of developing and maintaining integration points within the code outweighs the benefits, simpler models like data integration might be preferable.
   - Example: A small business only needs to sync customer lists between two systems rather than implementing a complex functional integration solution.
4. **Only static or infrequent data exchange is required**
   - If systems only need to exchange data periodically instead of interacting in real time, data integration may be a simpler solution.
   - Example: A company only needs to export daily sales reports from the sales system to the accounting system.

Functional Model Advantages

- **Comprehensive access**: Provides access to both data and business logic of the systems.
- **High customization**: Allows highly flexible and tailored integration solutions.
- **Legacy support**: Can update or extend legacy applications when needed.
- **Data consistency**: Ensures synchronized and consistent data across the system.
- **Versatility**: Can also address presentation or data integration issues.

Functional Model Disadvantages

- **Complex implementation**: The most difficult integration model to design and implement.
- **Higher maintenance cost**: More integration points in the code lead to higher maintenance effort.
- **Risk of data inconsistency**: Poorly designed integration can result in data mismatches.
- **Impact on legacy applications**: Integration may introduce risks to older systems.
- **Limited access to business logic**: Some applications may not provide source code or APIs, making it difficult to integrate business logic.

Heuristic 2

Build and maintain options as long as possible in the design and implementation of complex systems. You will need them.

Heuristic 3

Simplify, Simplify, Simplify

1. Using a table, Comparing and Contrasting the Three Approaches of Functional Integration Approaches?
2. Using a table, show the advantages and disadvantages of the 03 different integration model.

Summary

- Three types of integration models:
  - Presentation Model
  - Data Model
  - Functional Model
- These models do not constitute a rigid integration design model, rather many integrated systems are a combination of styles
- Each model has advantages and disadvantages
- The integration style should be selected based on the user requirements and constraints
- Think about the heuristics when selecting an integration style

The end

THANK YOU